

Phase 3: Project Design Phase — Proposed Solution Description

Project Name: Cab Booking (UCab)

Project ID: N/A (Solo Track Submission)

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1. Comprehensive Solution Overview

UCab is an end-to-end full-stack web platform built on the MERN stack (MongoDB, Express.js, React.js, Node.js) that transforms how urban commuters book cab rides and how fleet operators dispatch vehicles. Unlike standard ride-booking clones that merely list generic vehicles, UCab introduces a multi-tier vehicle categorization system coupled with a transparent deterministic fare calculation engine, live GPS state tracking, and a robust dual-engine persistence layer.

2. Key Pillars of the Proposed Solution

Pillar 1: Multi-Category Fleet & Transparent Fare Calculation

Instead of a single generic cab rate, UCab categorizes its inventory into four clear vehicle classes:

- Mini (10/km): Affordable hatchbacks designed for short solo commutes (Base: 140, 4 seats).
- Sedan (12/km): Executive sedans offering extra legroom (Base: 150, 4 seats).
- SUV (18/km): Multi-utility vehicles for family and group travel (Base: 180, 6 seats).
- Luxury (35/km): High-end business-class sedans (Base: 150, 4 seats).

When a user selects their pickup and drop-off points, the frontend instantly computes the route distance (km) and displays the exact fare breakdown:

Total Fare = Base Fare + (Distance * Price Per KM) - Promo Discount + Perks

Pillar 2: In-Ride Customization & Social Responsibility Suite

UCab enables riders to personalize their journey prior to boarding:

- 0- In-Ride Refreshments Package (+ 150): Automatically instructs the assigned driver to keep chilled mineral water and premium snacks ready inside the cabin.
- 0- Road Safety & Driver Welfare Donation (+ 120): Empowers riders to round up their fare, contributing directly to an emergency healthcare and accident insurance fund for drivers.

Pillar 3: Real-Time Dispatch Tracking & Corporate Invoicing

Live Status Tracker (UserHome.jsx): Once a booking is confirmed, the user experiences a visual 4-stage tracking indicator (Booking Accepted ! Driver Assigned ! Cab in Transit ! Ride Completed) accompanied by complete driver credentials (Vikram Sharma - 4.9 &, contact phone, and vehicle registration plate).

QR Corporate Receipts (ReceiptModal.jsx): Commuters can visit their booking history at any time (MyBookings.jsx) and download formal thermal corporate invoices complete with an embedded QR Verification Code for seamless company tax expense reimbursement.

Pillar 4: Executive Admin Control & Fleet CRUD

For dispatch leads and fleet owners, UCab provides a centralized command dashboard (AdminHome.jsx) that displays live key performance indicators (Total Gross Revenue 1, Total Bookings, Active Users, and Operational Fleet Count). Administrators can effortlessly add new cab inventory (AddCar.jsx), edit pricing and number plates (EditCar.jsx), toggle vehicle operational status between Available and Busy, and update trip dispatch states (AdminBookings.jsx) across the platform.

Pillar 5: Zero-Config Dual-Engine Database Architecture

To guarantee that the application can be evaluated or demonstrated on any laptop or university lab workstation without external database dependencies, a self-contained persistence engine (`server/db/store.js`) was engineered. If the Express server detects that MongoDB is offline on startup, it automatically switches from Mongoose ORM queries to atomic file operations (`data.json`), ensuring zero crashes and 100% data preservation.